

Streaming Instability under Forced External Turbulence: Feasibility and Robustness



Planet Formation in the Southwest, 3rd Meeting
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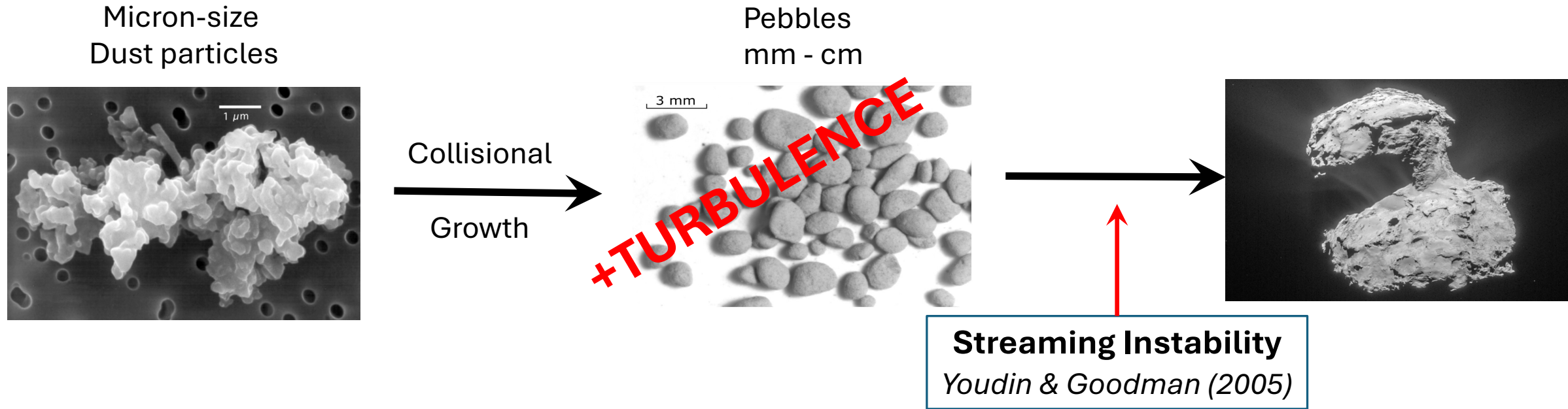
Wladimir Lyra

Paul R. Estrada

Jeffrey N. Cuzzi



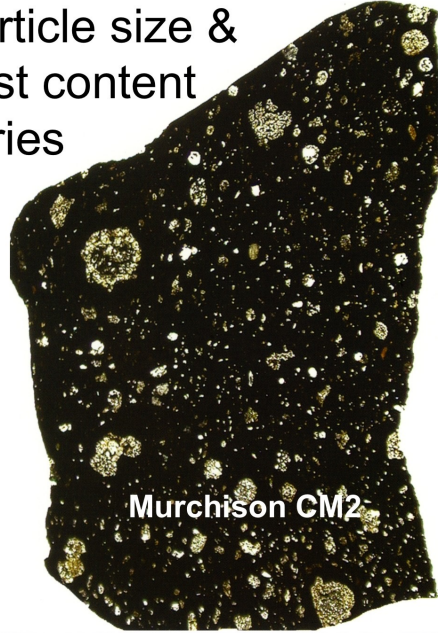
Planetesimal Formation: The Problem



Streaming Instability is, so far, the most successful mechanism for the formation of these ~100 km size objects.

Message of the Meteorites: The Ground Truth

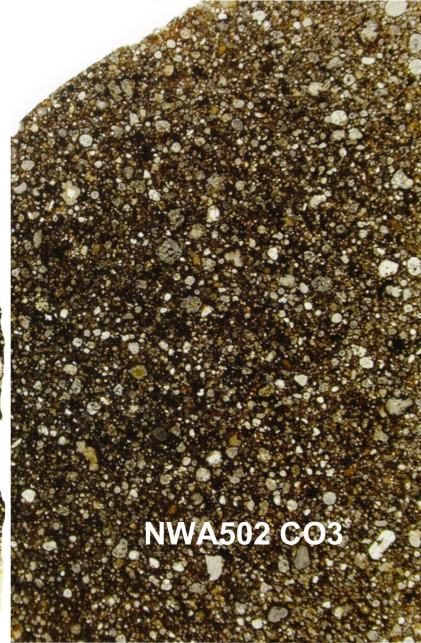
Particle size &
dust content
varies



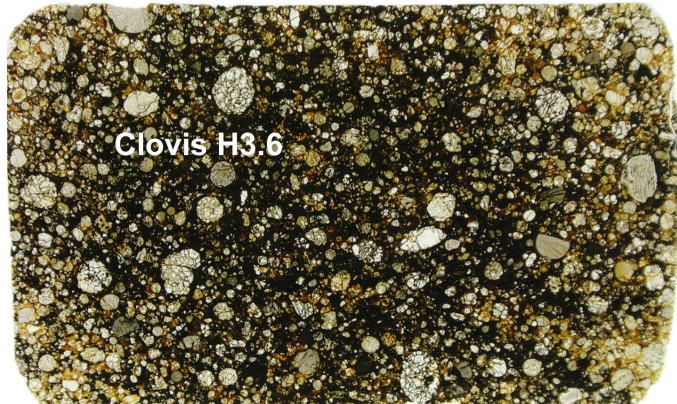
Murchison CM2



Vigarano CV3

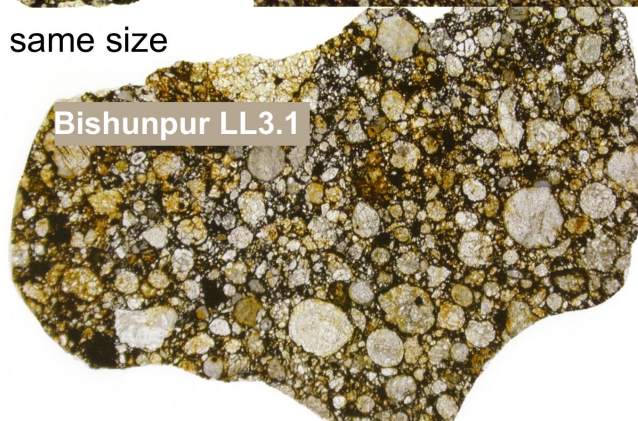


NWA502 CO3



Clovis H3.6

All same size



Bishunpur LL3.1

- Chondrules
- Chondrule aggregates
- Fine grain matrix

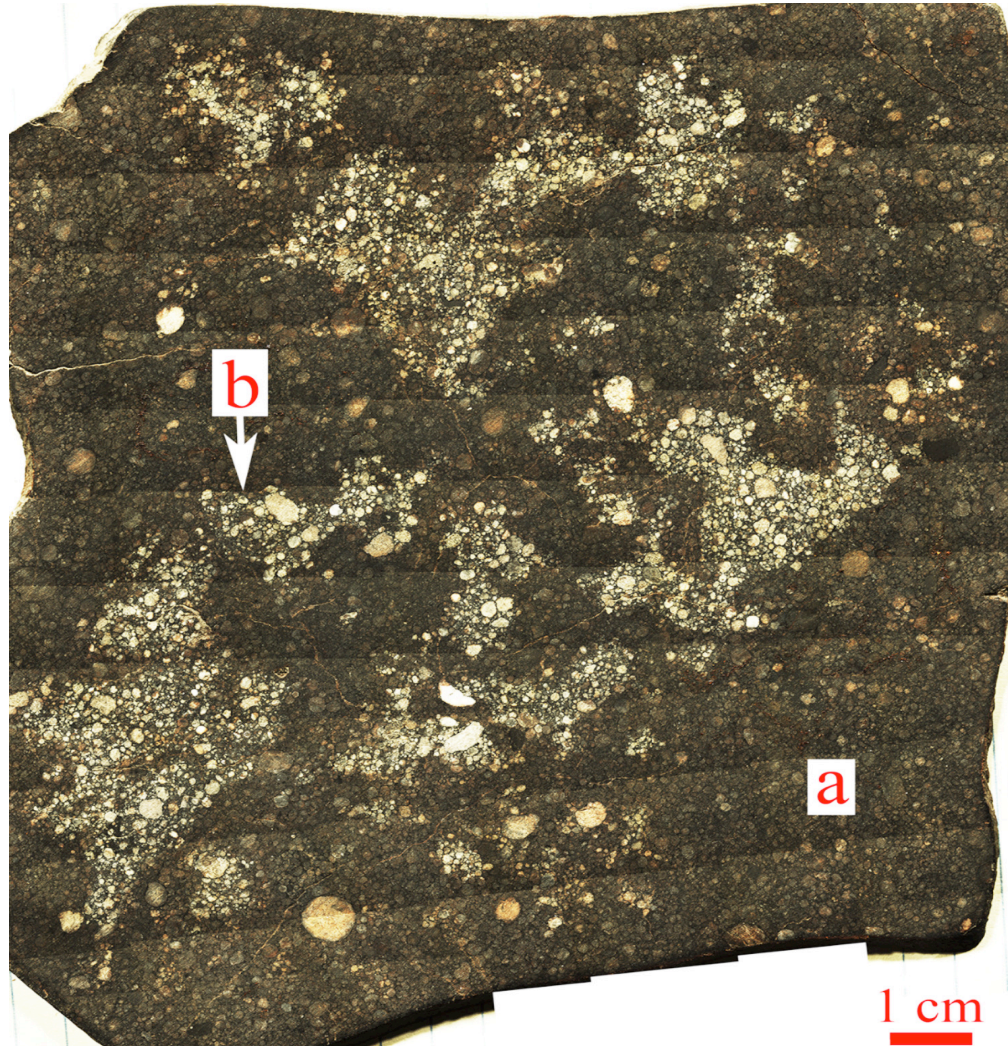
** Individual chondrule
size ~ mm or less
** Up to 80% by mass in
chondritic meteorites

Simon et al. (2018) EPSL

Take-away: Reasonable particle
sizes are a few mm to a few cm.

Inferring Turbulence from Meteorites.

NWA 5717



Chondrules in the two lithologies:

(a) Different Mg-Fe chemistry

(b) Different O-isotopes

(c) Different size distributions

(Bunch+2010, Cato+2017, Simon+2018)

Take Away:

Chondrules formed and accreted into aggregates in two different environments, before mixing together at a location where they were incorporated into a planetesimal.

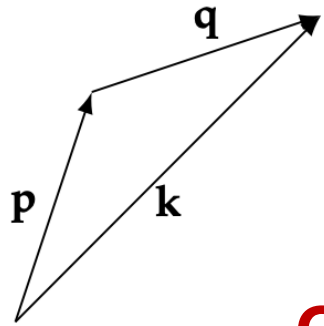
⇒ Presence of global turbulence

Theoretical, Numerical & Observational Evidences.

Reynolds Number (Re)

$Re \sim 10^{18}$ in a planet forming region.

$$(\partial_t + \nu k^2) \hat{u}_i(\mathbf{k}) = -\frac{i}{2} \mathcal{P}_i^{jl}(\mathbf{k}) \int d\mathbf{p} d\mathbf{q} \delta(\mathbf{p} + \mathbf{q} - \mathbf{k}) \hat{u}_j(\mathbf{p}) \hat{u}_l(\mathbf{q}).$$



$$\mathbf{k} = \mathbf{p} + \mathbf{q}$$

Plenty of modes and ample time for mode interaction (Triad)

Observation

Disk Observations estimate

$$\alpha \sim 10^{-4} - 10^{-3}$$

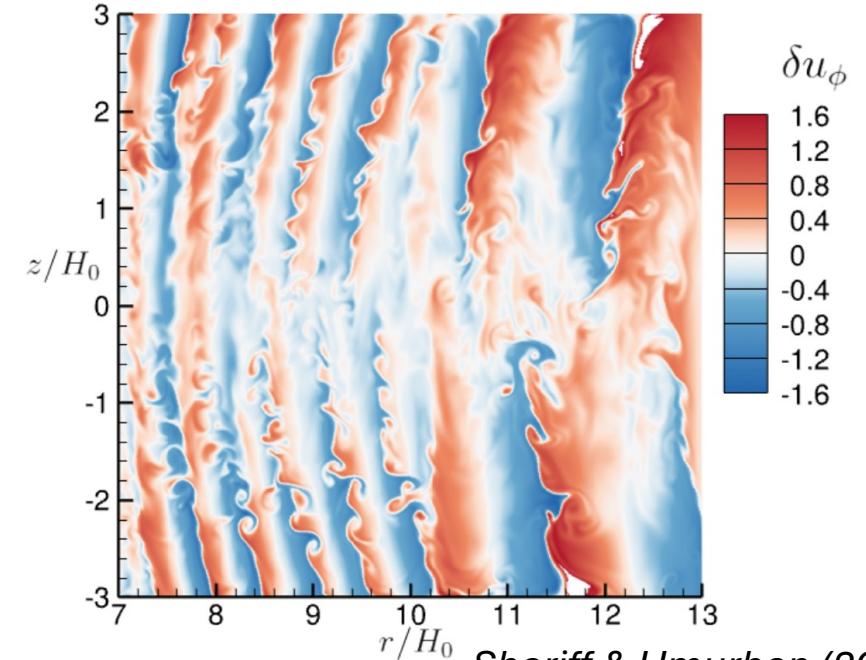
Flaherty et al. (2017, 2018, 2020)

Teague et al. (2018);

Doi & Kataoka (2021, 2023)

Villenave et al. (2022, 2023)

Simulation



Shariff & Umurhan (2024)

**Various known Instabilities:
MRI, VSI, COV, ZVI and the ones
yet to be discovered.**

Turner et al. (2014); Lyra & Umurhan (2019)

Streaming Instability: Conditions

1: Particle Sizes: mm – cm pebbles

$$\text{St} \sim 10^{-3} - 10^{-2}$$

2: Level of turbulence: Moderate

$$\alpha \sim \text{a few} \times 10^{-4}$$

Condition for SI to be triggered at the midplane:

Midplane dust-to-gas ratio $\mathcal{O}(1)$ as originally demonstrated by *Youdin & Goodman (2005)*

Later re-established by *Umurhan et al. (2020) (for turbulence)*

Most SI simulations are executed under laminar conditions, except:
Johansen et al. (2007), Gole et al. (2020), Lim et al. (2024) and Eriksson et al. (2025)

Implementation of Turbulence by External Forcing

Forcing vs Instability

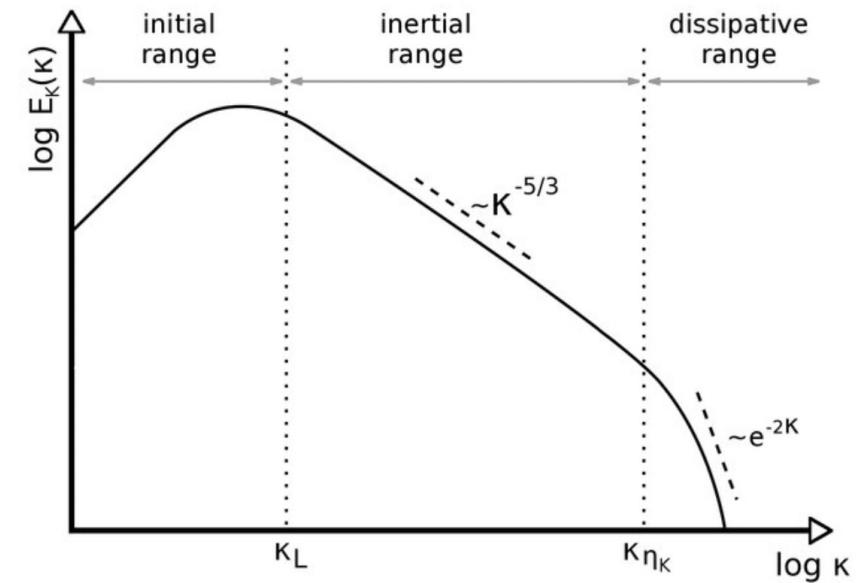
Why is forcing so popular in the field of turbulence? Why is forced turbulence adequate for particle clustering?

The answer lies in the definition of inertial range:

In a turbulent flow with sufficiently large Reynolds number, the multi-point probability distributions of small-scale relative velocities in the **inertial subrange** are uniquely determined by the **average rate of viscous dissipation** (ε), and are independent of viscosity.

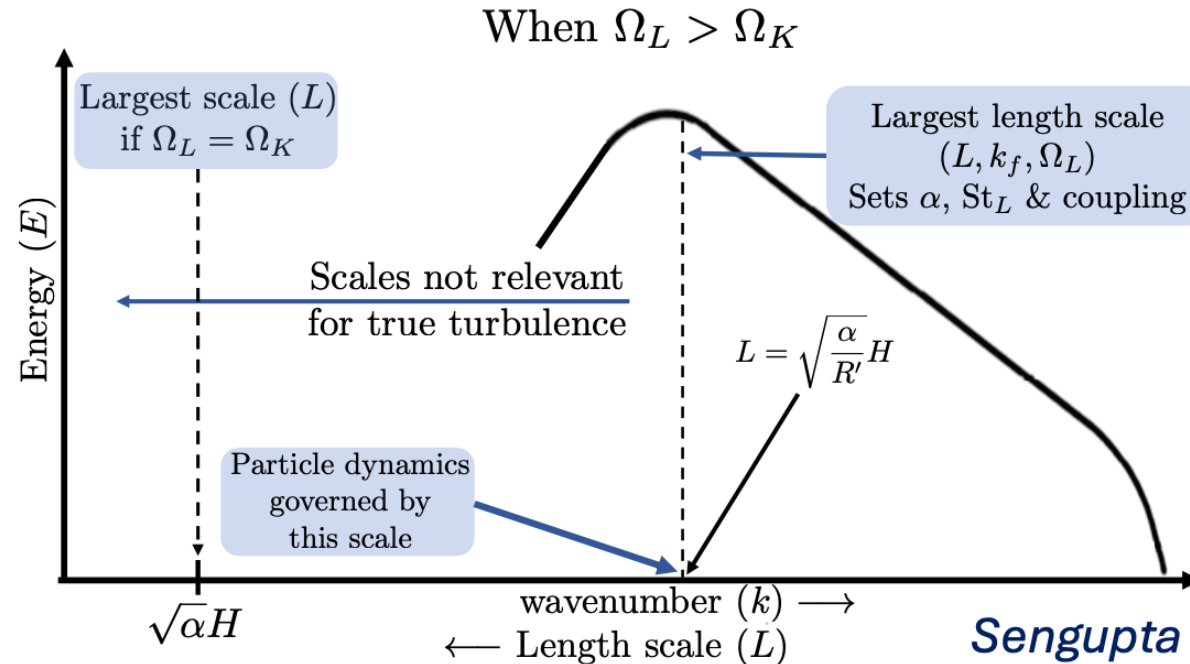
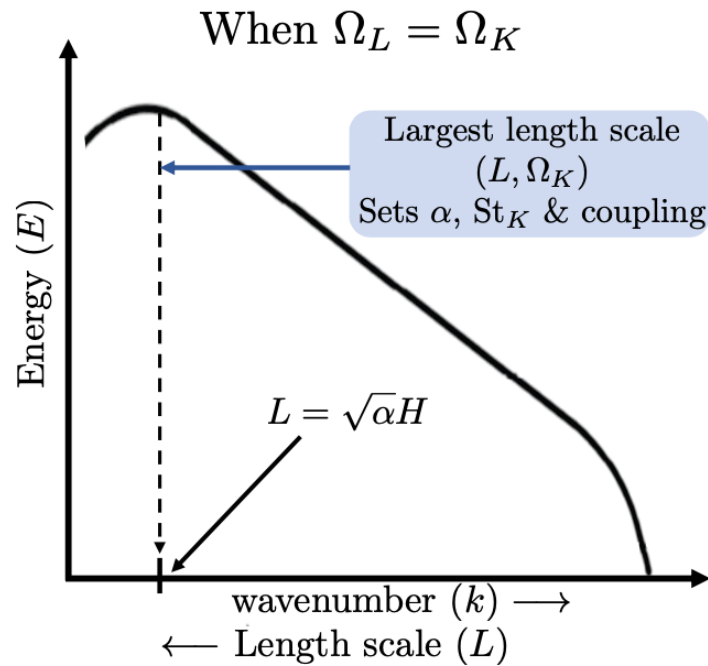
Instabilities affect the large (driving) scales --> Geometry of the system gets Imprinted at large scales --> Memory is lost at the inertial range.

A typical turbulence spectra



Forcing and Turbulence: The Case of Large Eddy

Mixing length formalism: Diffusion $\sim$ (some length scale) $\times$ (some velocity scale)
 $(D, \nu) \sim \alpha c H$



Sengupta et al. (2024) ApJ

$\tilde{\alpha} = U_{\text{rms}}^2 / c^2$: Uncorrected

$\alpha = \tilde{\alpha} / R'$ (S+24 Scaling)

$$R' = \Omega_L / \Omega_K = \sqrt{2k_f^3 E(k_f) / \Omega_K}$$

$$= 2 \times \text{Local Rossby Number}$$

St_K , St_L , St_ℓ and St_η

Stokes Number $St = (\text{friction time}) \times (1 / \text{some time-scale})$

$St_K = \tau_s \Omega_K$: Decides particle behavior wrt disk dynamics

$St_L = \tau_s \Omega_L$: Decides particle behavior wrt turbulence

$St_\ell = \tau_s \Omega_\ell$: Stokes number for eddies in inertial range

$St_\eta = \tau_s \Omega_\eta$: Stokes number wrt Kolmogorov scale

$$St_L = R' St_K$$

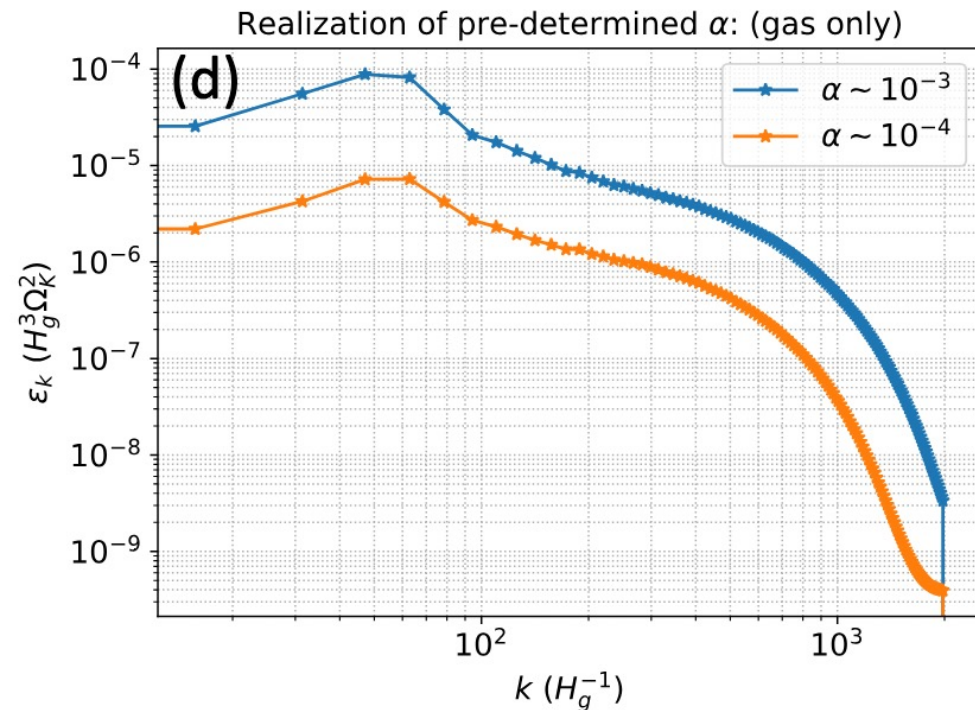
Simulation Setup

Standard SI Models:

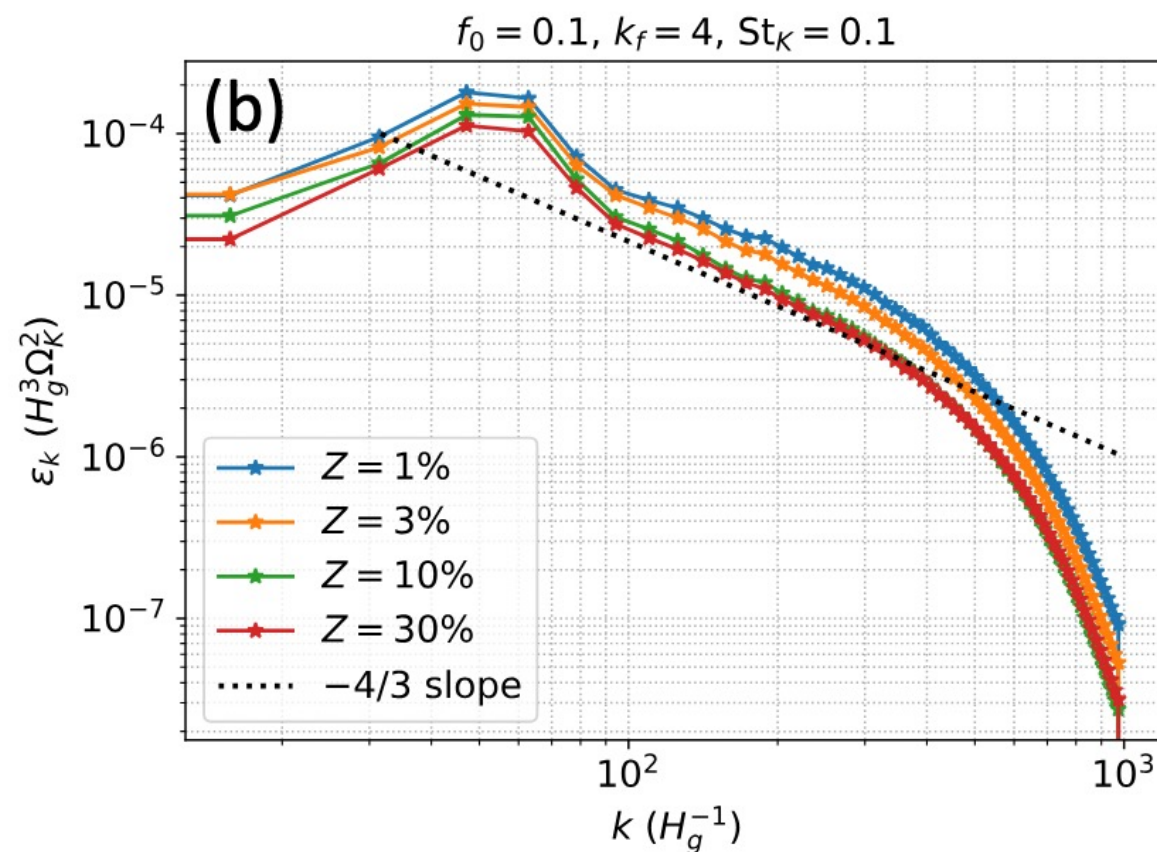
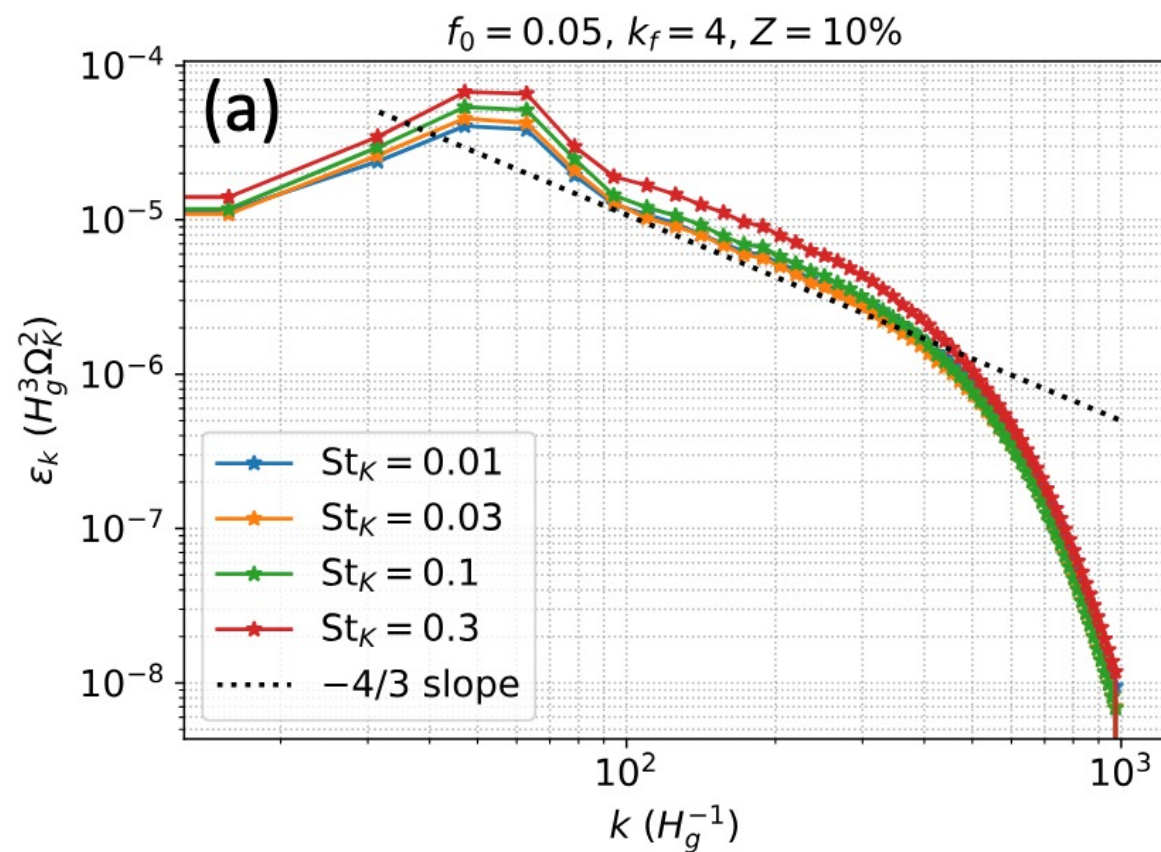
- Radial Pressure Gradient: 0.05
- Cubic box with side $0.4 H_g$, Periodic B.C.
- Grid size: 256^3 or 512^3 (depending on SI length scale)
- Forcing at wavenumber $k = 4$ (Delta correlated, isotropic)

Energy Spectra for $\alpha = 1e-4$ and $1e-3$

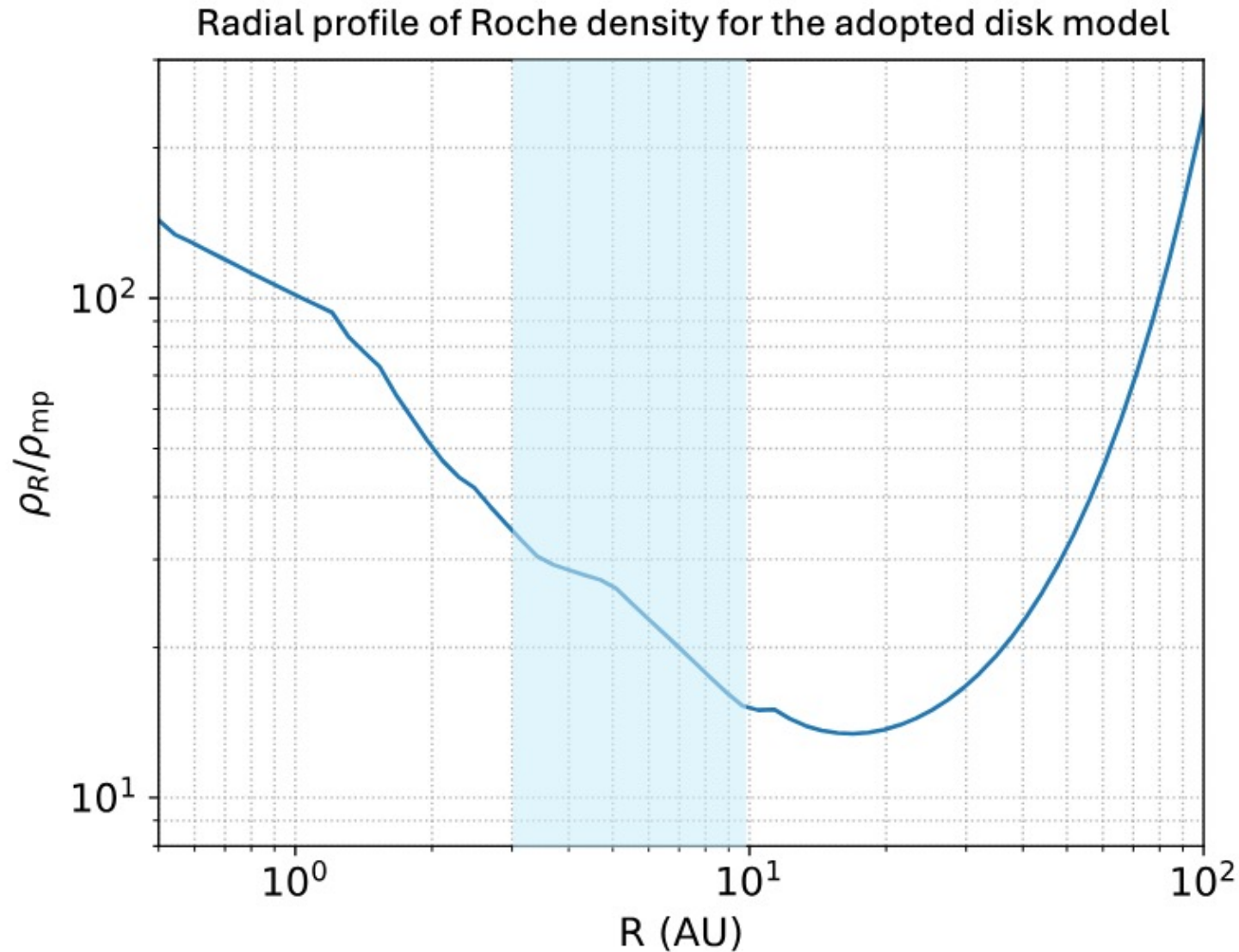
Wavenumbers are normalized by domain size



Effect of Mass Loading: Setting the correct α



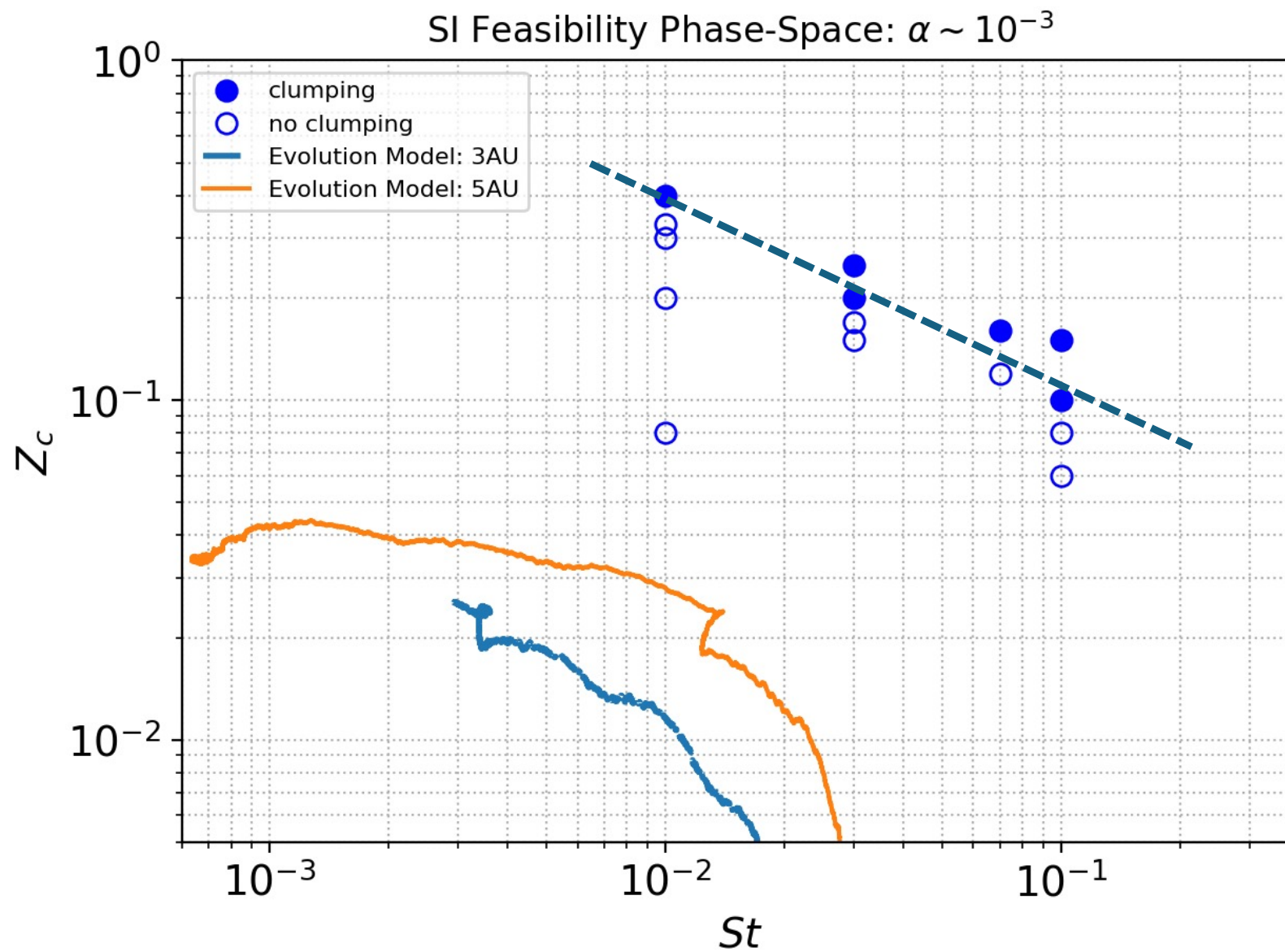
Criteria for Overdensities



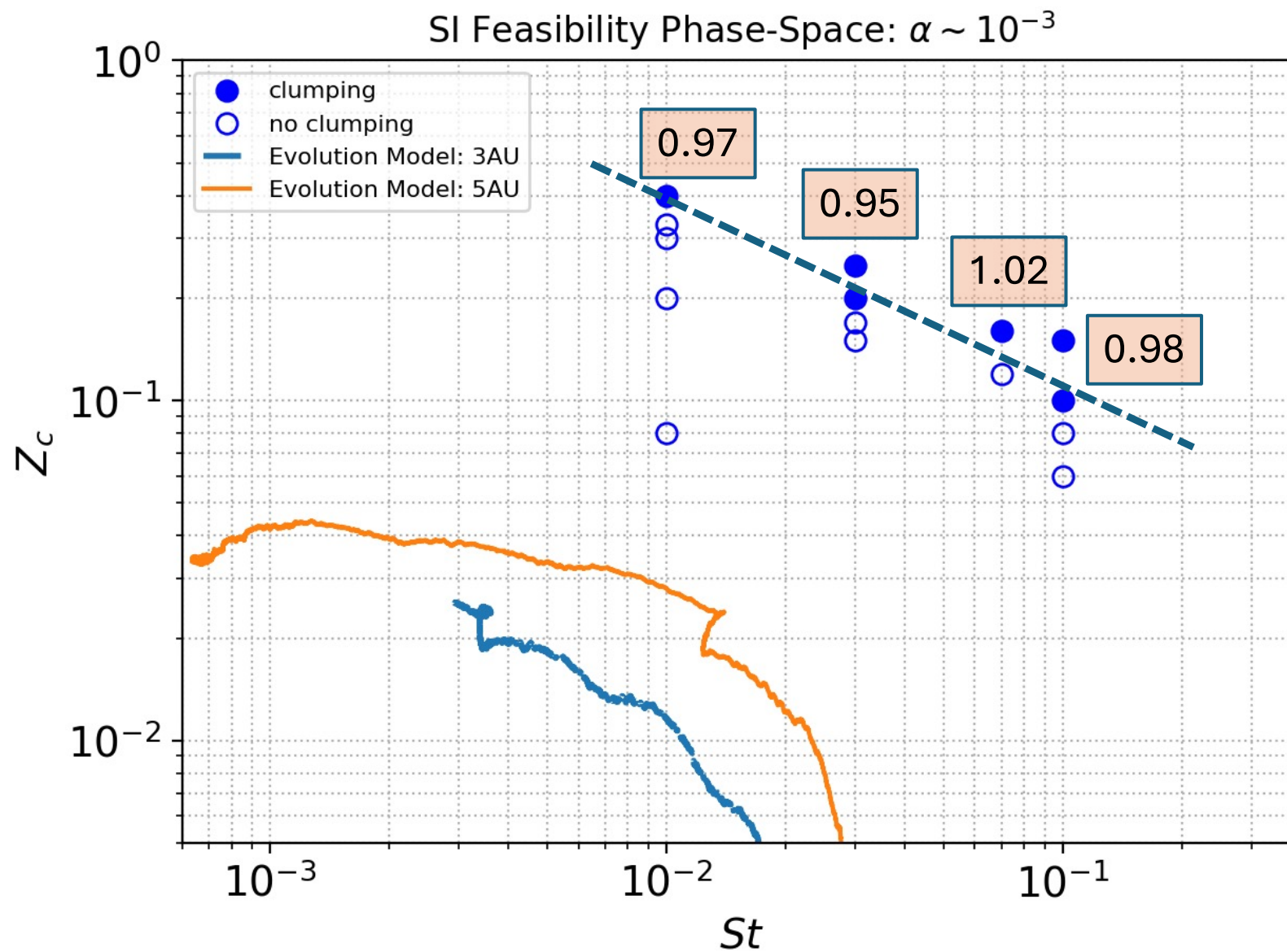
Disk model adopted from
Estrada et al. (2016),
Sengupta et al. (2022)

We adopt ~ 2 times the Roche
Density as the criteria for
Planetesimal formation.

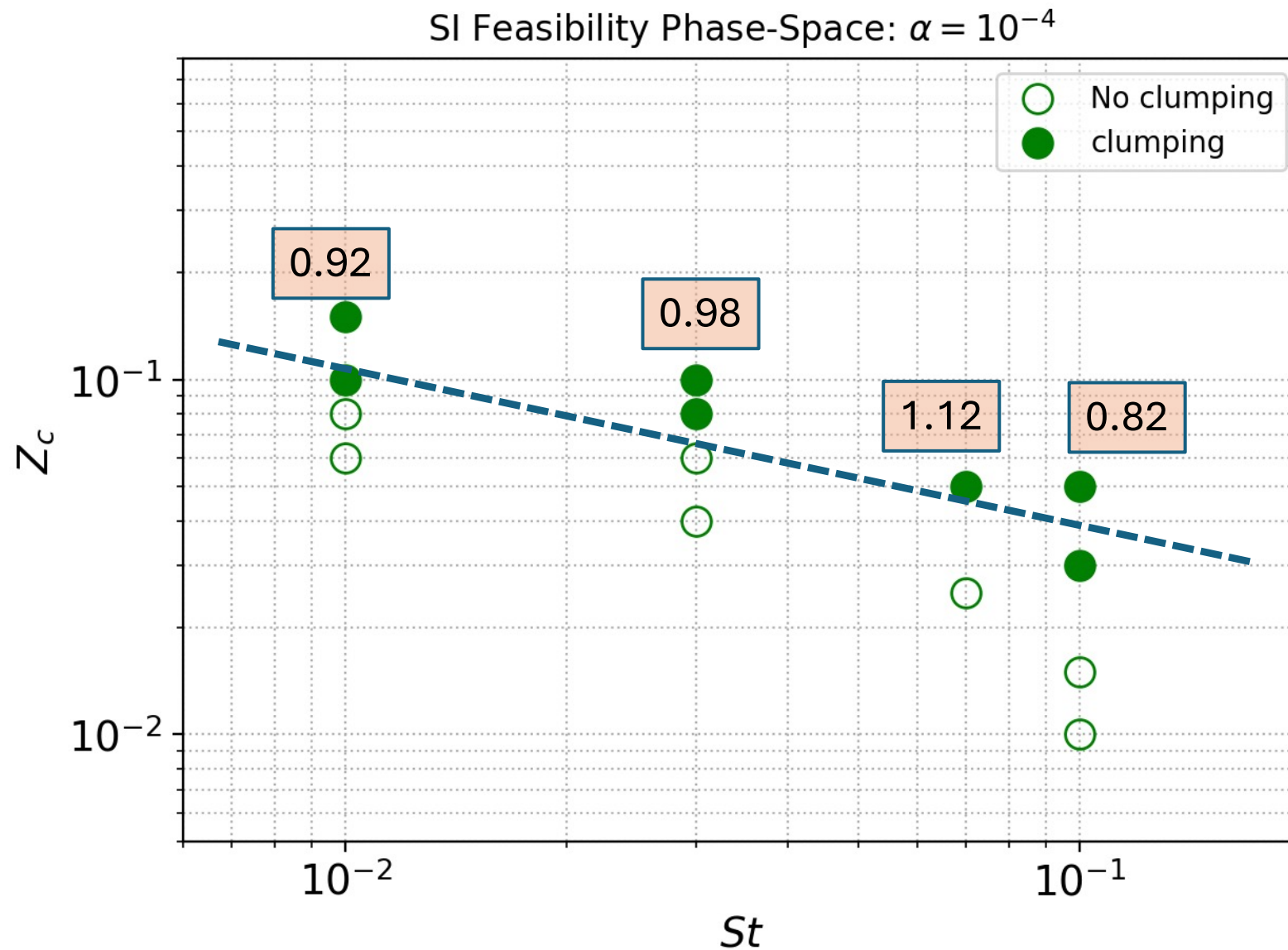
SI Feasibility Criteria



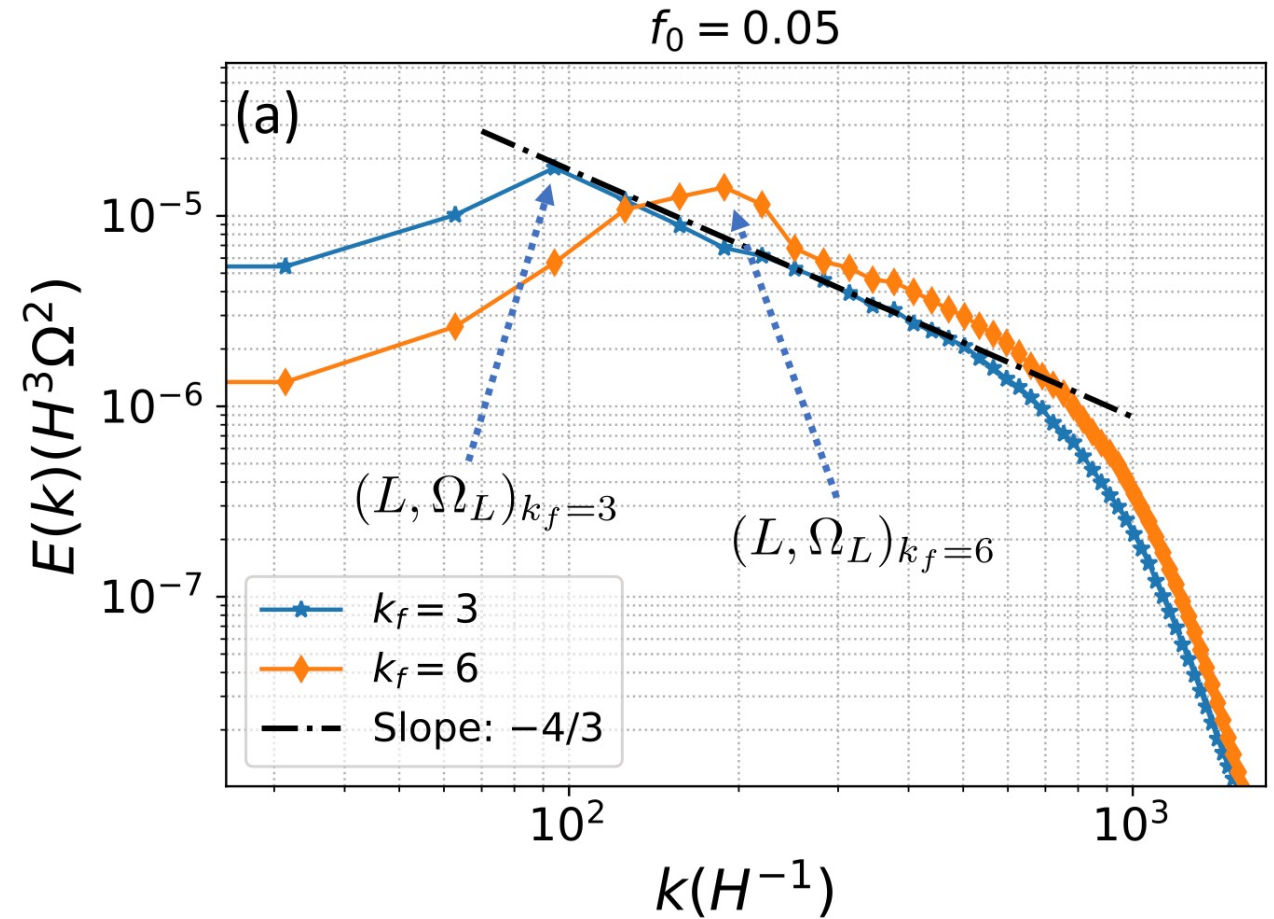
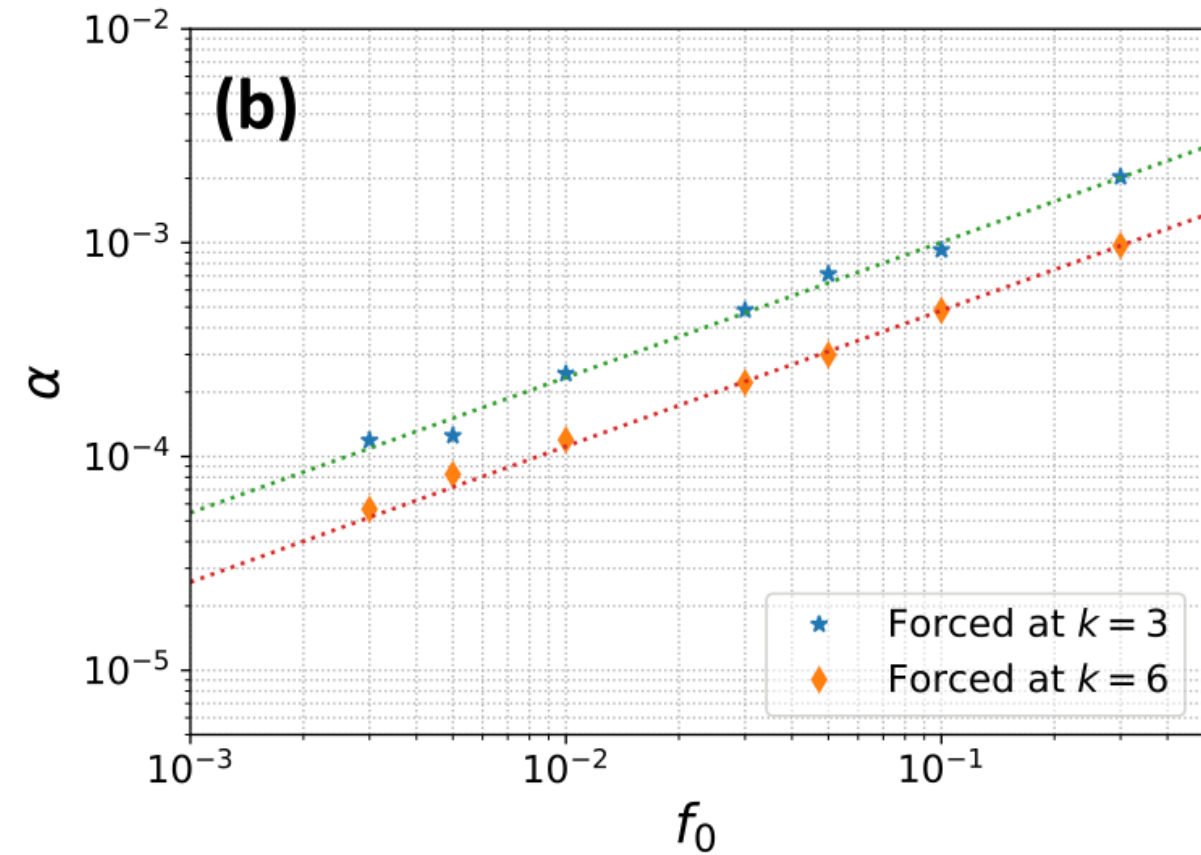
SI Feasibility Criteria $\epsilon_{mp} \sim 1$



SI Feasibility Criteria $\alpha \sim 10^{-4}$



α : Not a Good predictor of Particle dynamics



α : Not a Good predictor of Particle dynamics

